Laboratory-Processed Sorghum-Based Alcoholic Beverage does not Alter Liver Function in Adult Male Wistar Rats

[*1]Oyakhare M.O., [1]Ikhajiangbe H.I., [2]Adie P.U., [3]Ogboh C.B.

[1]Department of Physiology, College of Medicine, Ambrose Alli University, Ekpoma, Nigeria.
[2]Department of Human Physiology, Faculty of Basic Medical Sciences, University of Calabar, Calaba, Nigeria.
[3]Oral and Maxillofacial Department, University of Benin Teaching Hospital, Benin City, Nigeria.

ABSTRACT

This study investigates changes in selected liver function parameters following 14 days' ingestion of laboratory processed sorghum based alcoholic beverage (*Burukutu*) in adult male Wistar rats. Thirty (30) male Wistar rats weighing 80-100g were divided into 6 groups with group 1 as control, groups 2 and 3 received 5% of *Burukutu* every other day and on weekend respectively while groups 4, 5 and 6 received 1%, 5% and 25% of *Burukutu* daily. These treatments lasted for 14 days. At the end of treatment, blood sample was collected for assaying serum proteins and enzymes levels herein reported. The data was subjected to One Way Analysis of Varian (ANOVA) and $p \leq 0.05$ was considered significant. The results showed that serum total protein, albumin, globulin, and liver enzymes; aspartate aminotransferase (AST), alanine aminotransferase (ALT) and alkaline phosphatase (ALP) activities in rats administered *Burukutu* were not significantly different compared with the control. However, these liver proteins and enzymes increased in a dose dependent manner and are likely increased following every other day ingestion compared to weekend ingestions. The findings suggest that ingestion of laboratory processed sorghum based alcoholic beverage for 14 days may not affect the functionality of the liver in adult male Wistar rats. However, higher doses for a long duration is likely to result in a significant impact on the liver functionality and thus, the need for further studies.

Keywords: Alcoholic beverage, Sorghum, Burukutu, Liver enzymes, Wistar rats.

***Correspondance:** oyakhireosenimusa@aauekpoma.edu.ng Tel.: +2348060346220

1. INTRODUCTION

In Nigeria, and many African settings, alcohol is culturally tolerated during festivities and ceremonies, especially in the absence of any religious restrictions [1]. Recent reports, however, have shown increased and alarming daily consumption of alcohol among adolescents and young adults, with Nigeria now ranked among the leading countries with the highest per capita consumption of alcohol globally [2]. Consequently, there have been increases in several related social and health issues including alcohol dependence, domestic violence, traffic injuries, and several chronic diseases, suggesting a need to develop more comprehensive population-wide preventive measures [1, 3, 4]. Alcohol use and misuse and related problems vary around the world and is substantial in several countries [5]. In 2016, the World Health Organization (WHO) reported that more than 3 million people died worldwide as a result of harmful use of alcohol, accounting for 1 in 20 global deaths and representing the world's third largest risk for disease and disability [5]. In specific terms, harmful use of alcohol accounted for 28.7% of deaths due to injuries, 21.3% of digestive diseases, 19% of cardiovascular diseases, and 12.6% of cancers globally [5, 6].

Developing countries like Nigeria depend mostly on indigenous technology for food

preparations especially food of plant origin including alcoholic beverages made from cereal grains [7]. Cereals have been malted for centuries for weaning of foods and brewing traditional alcoholic and non- alcoholic opaque beverages in Africa long before the advent of Western style malting and brewing technologies [8]. In Africa's developmental history some beverages have evolved with the people and served as their cultural identity and socializing drinks. African traditional beverages are known by different names in different parts of Africa. These names include *Ikigage* in Rwanda, *Tchoukoutous* in Benin and Togo, *Dolo* in Burkina Faso, *Burukutu* and *Pito* in Nigeria and Ghana, *Amgba* in Cameroon, *Doro* and *chibuku* in Zimbabwe, *Bill bili* in Chad, *Kaffi* in South Africa, *Bouza* in Ethiopia, *Pombe* or *Bwalwa* in Tanzania, *Bejalwa ja Setswana* in Botswana, and *Yalwa joule* in Lesotho [8, 9]. These alcoholic beverages are enjoyed for pleasure shortly after their production [10] and its production and sale serve as a source of income for many [7].

Of interest in this study is the popular known sorghum based alcoholic beverage known as *Burukutu. Burukutu* is a brown coloured alcoholic beverage brewed traditionally by Africans which has the flavour of vinegar [11] with an opaque colour that is due to the presence of yeast materials and suspended solid [12, 13]. *Burukutu* is a locally brewed nutritious alcoholic drink with a vinegar-like taste and a cloudy appearance that is primarily made from the grains of either *Sorghum bicolor* or *Sorghum vulgare* which are rich in energy and protein compounds [9, 14-17]. It is consumed by an estimated population of 10 million people in West Africa and both male and female of all ages including children and pregnant women consume it [18].

In Nigeria, *Burukutu* has become a choice drink across the country and now popularly used in serving guests during ceremonies and festivals and has a high patronage especially among impoverished individuals and low income earners because of its affordability [11]. The alcoholic content ranges between 3-6% [12, 19] and has a short shelf life (ranging between 1-8 days) attributed to low titratable acidity, lactic acid and alcohol content, high vitamins and presence of lipoxidation products [15, 20]. However, its spoilage can be delayed for two weeks if the product is pasteurized [13]. Disappointingly, delaying spoilage promotes starch gelatinization in sorghum beer which increases its viscosity, cause the removal of amylolytic enzymes and active yeasts resulting in poor foaminess [21]. European barley beers generally have less nutritional value compared with sorghum beer as a result of high yeast content, lactic acid bacteria and other suspended materials [22]. Brewing of *burukutu* involves lactic acid fermentation unlike European beer [21]. Lactic acid always present in burukutu and it low pH (ranging between 3.3-3.5) is responsible for its sour taste [23, 24].

Traditionally, the preparation of *burukutu* involves several stages using mainly starchy grains of guinea corn, millet and maize [24, 25] which are largely produced in households usually under unhygienic conditions that predisposes the product to microbial contamination [26]. Most often, *burukutu* is sold and consumed in a microbiologically active state within a highly populated vicinity where its production takes place [18]. In recent times, many researchers have assessed the microbiological quality, mineral contents, physicochemical properties and proximate composition of *burukutu* sold in different towns and localities [13, 15, 19, 20]. Currently, there is limited studies which compared the quality of *burukutu* sold in various locations and a similar product brewed in the laboratory using standard procedures [22]. Therefore, this study investigates the changes of some serum liver function biochemical variables in response to ingestion of *Burukutu* in an attempt to elucidate further the impact of locally made sorghum-based alcoholic beverage on the functionality of the liver in adult male Wistar rats. Specifically, the objectives of the study include to assess and compare serum total

protein, albumin, globulin levels and serum aspartate aminotransferase (AST), alanine aminotransferase (ALT) and alkaline phosphatase (ALP) activities between test and control.

2. MATERIALS AND METHODS

2.1. Materials: Sorghum (*Sorghum bicolor*) was locally purchased from Hausa Quarters in Ekpoma market square; the local market in Ekpoma, Edo State. The animals (adult male Wistar rats) were obtained from the Animal Holding Unit of the Faculty of Basic Medicial Sciences, Ambrose Alli University, Ekpoma. Rat chow (grower's mash) was palletised chicken feed (Grower) produced by Grand Cereals Ltd, Jos, Plateau State, Nigeria and was purchased in Ekpoma, Edo State, Nigeria.

The instruments including dissecting set, magnifying lens, weighing balance, centrifuge used in this study were obtained from the Department of Physiology, College of Medicine, Ambrose Alli University, Ekpoma, Edo State-Nigeria. Others such as plastic basket, Wattman No. 1 filter paper, orogastric cannula, plates for food and water, syringes, centrifuge bottles and hand gloves were obtained from a local store in and around Edo State, Nigeria. All other chemicals were obtained from a chemical store in Ekpoma and the solution needed prepared in the physiology laboratory where the experiments were conducted.

2.2. Preparation of the Sorghum based alcoholic beverage: The production of Burukutu in this study was done following steps categorized broadly into malting, fermentation of mash and the maturation process as in the method described previously by Wedajo Lemi [27] with few modifications. The purchased sorghum grains (2kg) were steeped in water overnight. The grains were then poured into a basket and the water allowed to drain off for about an hour. The grains were then spread out on a mats and covered with layer of leaves. During this malting period (3 days), the grains were occasionally turned over. Germination, which starts within 24 hours after steeping, is allowed to continue for this period. After germination the malt was spread in thin layers in the sun to dry for 1 day. The dried malt was grounded into powder. In the next stage, the ground malt was mixed with water and stirred with a stick. The resulting mixture was left to ferment for 2 days and thereafter the mixture was boiled for about 4 h. The boiled mixture was then filtered and left to mature for another period of 2 days. The end product was a suspension of some particles in a creamy liquid and this was left to stand for 24 hours and the water decanted and evaporated to a semi-solid.

2.3. Experimental Animals: Thirty male Wistar rats (N = 30) of comparable weight (100±20g) were used for the study. They were given standard care of laboratory animals as stated in the guidelines and regulations for the use of animals in scientific experiment. They were housed in a plastic cage (60 cm x 30 cm x 30 cm) with top ventilation under standard laboratory conditions in a well-ventilated room under a 12/12 hours light/dark cycle. They were allowed two weeks of acclimatization and fed rat chow and water *ad libitum.*

2.4. Experimental grouping: The animals were divided into 6 groups of 5 rats each as follows;

i. Group A serves as the control and received clean water and rat chow throughout the period of treatment.
ii. Group B (test group 1) received in addition to clean water and rat chow, 5% *burukutu* in their diet at the beginning of treatment and every other day for the 14-day study.
iii. Group C (test group 2) received in addition to clean water and rat chow, 5% *burukutu* in their diet at the beginning of treatment and every weekend (Fridays, Saturdays and Sundays) for the 14-day study.
iv. Group D (test group 3) received in addition to clean water and rat

chow, 1% *burukutu* daily in their diet for the 14-day study.

v. Group E (test group 4) received in addition to clean water and rat chow, 5% *burukutu* daily in their diet for the 14-day study.

vi. Group F (test group) received in addition to clean water and rat chow, 25% *burukutu* daily in their diet for the 14-day study.

2.5. Sample Collection: At the end of the treatment (14 days), the animals were cervically decapitated and blood sample collected following standard laboratory procedures. The animal was dissected along the lateral side of the neck to assess the jugular vessels. Blood sample was then obtained using 2mL syringe into plain bottles. The blood sample was centrifuge at 3000 rpm and the serum obtained for liver function evaluation using commercial kits (Randox Laboratories kits, United Kingdom). These parameters were estimated from serum samples as per the procedure described in the kit.

2.6. Sample Analysis: Aspartate Aminotransferase (AST) and Alanine aminotransferase (ALT) were determined using the method outlined in Randox Kit. Alkaline Phosphatase (ALP) was determined using the method outlined in Teco Kit.

2.7. Statistical Analysis: The Statistical Package for Social Sciences (SPSS version 20) was used for data analysis. The one-way analysis of variance (ANOVA) was employed for data analysis and where applicable LSD was determined and confidential interval of $p \leq 0.05$ considered statistically significant. Results were presented as mean ± Standard deviation using suitable tables and charts.

3. RESULTS

Compares serum liver proteins in *Burukutu* treated and control rats: Figures 1 presents the serum total protein, albumin and globulin levels in rats administered *Burukutu* and untreated control respectively. Serum total protein, albumin and globulin levels were not significantly different between groups compared with the control. However, serum total protein, albumin and globulin levels were observed to increase in a dose dependent manner from the 1% to 5% and 25% *Burukutu* treated groups. Also, serum total protein, albumin and globulin levels were more likely decreased in the weekend *Burukutu* treated group and more likely increased in every other day treated group compared with the control.

Compares serum liver enzymes activities in *Burukutu* treated and control rats: Figures 2 presents the serum AST, ALT and ALP activities in rats administered *Burukutu* and untreated control respectively. Serum AST, ALT and ALP activities were not significantly different in the *Burukutu* treated groups compared with the control. However, serum AST, ALT and ALP activities increased in a dose dependent manner in the 1%, 5% and 25% *Burukutu* treated groups. Also, serum AST, ALT and ALP activities were more likely increased in every other day *Burukutu* treated group and decreased in the weekend treated group compared with the control group.

4. DISCUSSION

The liver function indices are parameters for measuring the functional status of the liver [28]. The elevation of transaminases characterizes liver diseases and dysfunction and when liver enzymes found predominantly in the liver are elevated in the serum is a clear sign of damage to the hepatocyte cell membrane [29]. The hepatobiliary enzymes, like AST and ALT, convey information on hepatocyte injury, whereas bilirubin, albumin convey information on liver function [30].

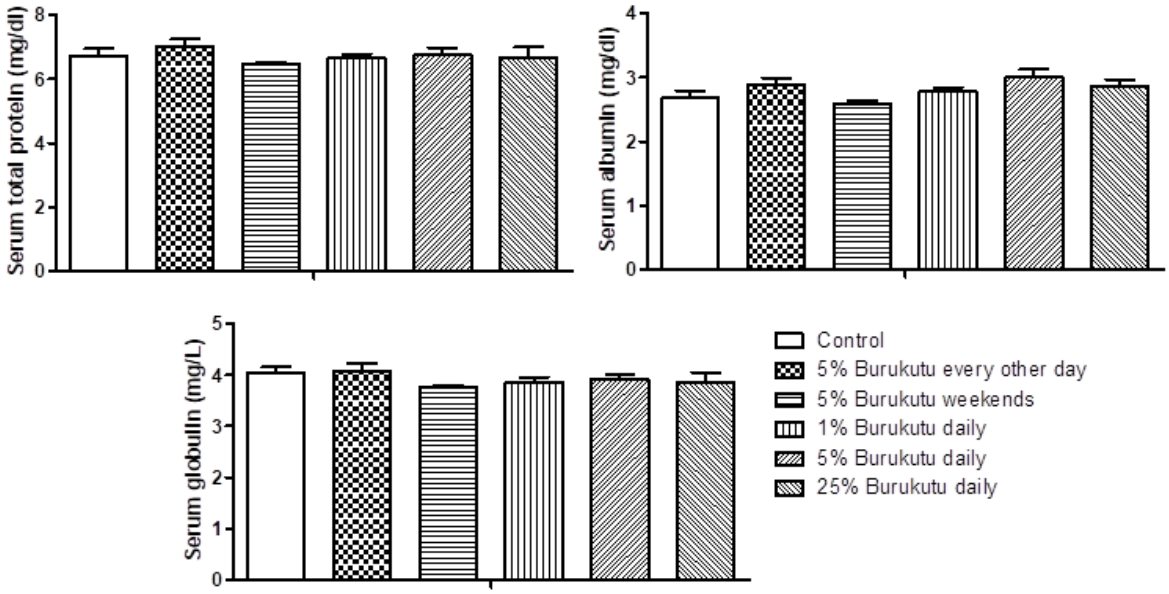


Figure 1. Serum total protein, albumin and globulin levels of *Burukutu* treated and untreated control in adult male Wistar rats

Values are means±SEM, No statistically significant different compared with control at p≤0.05.

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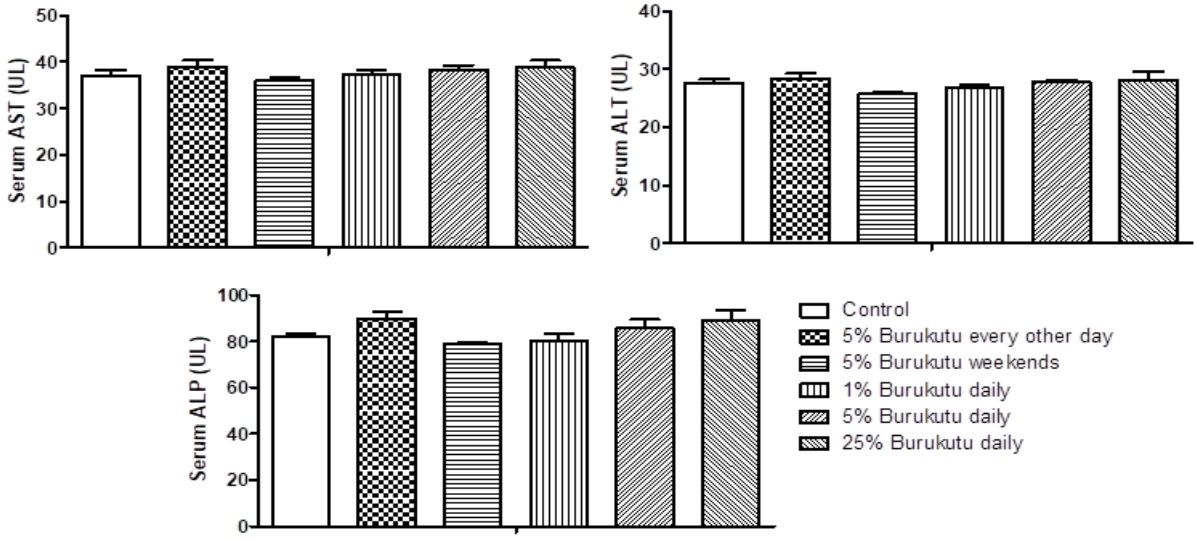


Figure 2. Serum AST, ALT and ALP activities of *Burukutu* treated and untreated control in adult male Wistar rats.

Values are means±SEM, No statistically significant different compared with control at p≤0.05.

Total protein, albumin and globulin are important protein produced by the liver [31]. Serum total and conjugated bilirubin represent the metabolic functions of the liver [32]. Decreased or increase in these liver protein may indicate an impairment to the synthetic function of the liver and could lead to severe cirrhosis, hepatitis, and fatty liver [33]. The serum levels of liver function parameters herein studied such as total protein, albumin and globulin following ingestion of *burukutu* reflected no significant different compared with the control group. This finding of no

alteration in total protein, albumin and globulin following ingestion of *burukutu* suggests that the alcoholic beverages did not alter liver function. The non-significant different observed in their levels in the groups treated with *burukutu* when compared with the control group may indicate that the ingestion of burukutu did not impair their production.

Elevated ALP, ALT and AST reflects impaired bile excretion, bile flow [32] and damage to the hepatocyte cell membrane [29]. From the results obtained, there was no significant difference in the activities of AST, ALP and ALT in the groups ingesting *burukutu* when compared with the control. This may indicate that *burukutu* do not affect or impaired bile excretion, bile flow, metabolic function and hepatocyte cell membrane of the liver since there was no elevation in these liver enzymes when compared with the control group.

Chronic alcohol consumption represents a major risk factor for the development of liver fibrosis, alcohol liver diseases (ALD), and hepatocellular carcinoma (HCC) [34-36]. In the present study however, laboratory processed *burukutu* did not alter liver function proteins and enzymes. In support of these findings, Ajiboye et al., [37] has reported burukutu consumption at 0.78 g/kg alcohol produced no significant change in the activities of ALP, ALT, and AST. In the same study, Ajiboye et al., [37] reported that at dosed of 3.71 and 7.43 g/kg *burukutu* resulted in significant decrease in the activities of ALP, ALT and AST in the liver tissue but with corresponding increase in serum. The non-significant alterations in the serum observed in this study may be that the alcohol content of the *burukutu* used in this study are below liver toxicity doses. Also, considering that *burukutu* is high in vitamins [15, 20] and other phytochemicals of the mainly starchy grains of guinea corn, millet and maize which [24, 25], it is likely that the non-toxic effect seen in this study are own to these nutritive contents. In addition, there has been report of lesser nutritional value in European barley beers compared with sorghum beer [22]. Thus the assertion that *burukutu* is a locally brewed nutritious alcoholic drinks [9, 13-16]. Hence, the observed comparable levels of serum liver proteins and enzymes in the treated and control rats in the present study.

5. CONCLUSION

The present study showed that liver function proteins and enzymes were not altered following 14 days of laboratory processed *Burukutu* ingestion in rats. However, the increased dose dependent changes in serum AST, ALT and ALP following daily ingestion of *burukutu* are the possibility of a dysfunctionality in liver physiological and metabolic function following higher doses and or chronic ingestion. Thus, monitoring and evaluation of serum proteins and enzymes are important following processed sorghum alcohol consumption. Based on the findings from this study, further investigation to assess liver function in longer duration of ingestion of processed sorghum based alcohol is recommended.

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